

Choosing The Right BENCH POWER SUPPLY



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The term bench brings about a mental image of a DC power supply used on an engineer's or a technician's bench for innumerable power tasks. These might be variable DC bench power supplies that appear to be relatively simple instruments, but engineers rely on these to deliver stable, precise and clean voltages and currents, regardless of load.

To identify the appropriate bench power supply for a particular application, one must come up with answers to several important questions and understand the basics of how power supplies are specified. We have to consider the following points to save substantial time and money later in the system configuration process.

Power, voltage and current requirements for device under test

Different types of power supplies can have different power envelopes (Figs 1a, b and c). In a rectangular power envelope type of supply, any load current can be supplied at any voltage level.

Another type has multiple rectangular envelopes for multiple ranges that offer the option of higher values of one parameter at the expense of the other.

Then there are those that can deliver a hyperbolic envelope that provides a more continuous transition than multi-range power supplies. In this kind of supply, one parameter is inversely proportional to the other.

High-power output supplies tend to have either a multi-range or a hyperbolic envelope. To make the right selection, take the time to evaluate the power levels required by the application.

Parameters to be considered

Although many key power supply parameters will vary depending on each application, following parameters are critical in all cases:

Accuracy. It determines how close the regulated parameter is to its theoretical value. Output uncertainty is largely due to error terms in case of a digital-to-analogue converter (DAC). Setting accuracy is tested by measuring the regulated variable with a traceable, precision measurement system connected to the power supply's output. It is given as $\pm$ (% of setting + offset).

Resolution. It is the smallest change in voltage or current settings that can be selected on the power supply. The resolution specification limits the number of settable discrete levels. A DAC with more bits produces finer resolution. But, with corrections for offset and gain errors, resolution will be less than the number of bits in the DAC. Setting resolution may be expressed as an absolute unit value or as a percentage of full scale.

Readback accuracy. It determines how close the internally-measured values are to the theoretical value of the out-

Fig. 1: (a) Versatile rectangular power envelope can supply any current to load at any voltage level, (b) multiple rectangular envelopes for multiple ranges and (c) hyperbolic envelope provides a more continuous transition than a multi-range power supply

